Statistical Foundations of Machine Learning

Homework 3

1. The discussions in Lectures 2-5 are mostly taken from the textbook by Ross. Specifically, the following is the coverage in the text.

   - Section 5.5 discusses the Normal random variable in some detail.
   - Section 6.2 discusses the sample mean and Section 6.3 the Central Limit Theorem. Section 6.4 is about the sample variance.
   - Section 7.2 discusses Maximum Likelihood Estimation.

   Read these chapters and make sure you understand the solved examples.

2. Practice problems. These are useful in understanding the concepts.

   (a) From Chapter 5 of the text, solve problems 31, 32, and 33.

   (b) From Chapter 6 of the text, solve problems 12, 14, 18, 19, 30. You will have to use the normal table for this.

   (c) From Chapter 7 of the text, solve problems 1, 3, and 5.

   Going forward, I will assume that you have solved the preceding problems and know the solutions and not just the answers!

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In this part, we describe computational problems to help visualise some of the concepts taught so far. It is expected that this exercise will take 60-90 minutes to get all the data and some additional time to put them all together into the submission.

1. Write a program to do the following:

   (a) Generate n i.i.d random variables of a specified distribution F whose mean and standard deviation are known.

   (b) Compute the sample mean and the limits of the 95% confidence interval.

   (c) Check if the true mean is in the 95% confidence interval;

   (d) In an outer loop, repeat this experiment K times.

   (e) Plot the histogram of the sample mean from the K experiments.

   (f) Represent the lower limit by a blue dot and the right limit by a red dot and make a scatter plot of both these limits.

   Once you have understood what you need to do, we will just change F, n, and K and see how things behave. This is next.

   (a) Let F correspond to Gaussian with $\mu = 0$ and $\sigma = 2$. Let $n = 20$ and $K = 1000$. Repeat for different $\sigma = 3, 5$ and $n = 50, 100, 200, 1000$.

   For each combination of σ and n, print the histogram and the scatter plot. Also mention the number of times that the true mean was inside the confidence interval calculate.

   (b) Now pretend σ is not known and use sample variance as discussed in class in all of the above.

   (c) Now repeat part 1 above for three cases of F being uniformly distributed: $(0, 1)$, $(-1, 1)$, and $(-2, 2)$.

   (d) Finally repeat for F being a Bernoulli distribution

   In your own words, in your own handwriting, and without asking an LLM describe what you see in the picture. Specifically, explain how the preceding gives you empirical illustration for LLNs, CLT, and of the properties of the sample mean. **I trust you to make an honest effort and make an honest submission. I would rather you make the effort and possibly be wrong than you submit an answer that you did not obtain yourself.**

   Submit the three sets of figures as per instructions and the handwritten writeup with your explanation.

2. Now consider the following.

   (a) A manufacturing line produces 100 ohm resistors; The buyer wants a tolerance of 5% and willing to accept batch if the manufacturer can guarantee that at most 1% of the components are outside the tolerance band.

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(b) Assume Gaussian distribution for the resistance values;

(c) Compute the σ that the process should have.

(d) Write a program to verify this, execute the program and generate the plots to empirically show that your choice of σ is correct.

Submit a short handwritten description of what you did. And provide the empirical evidence.

3. We will continue with the manufacturing line for resistors.

(a) An inspector is sampling from a large lot of, say, 1 million resistors.

(b) The inspector draws n samples and accurately measures their resistances.

(c) Assume that the resistances are i.i.d. Gaussian with unknown mean and variance

(d) Inspector wants to be *very sure* that less than one percent are outside the tolerance band.

(e) From the measurements, determine the $C\%$ confidence interval for μ and σ.

(f) Design the acceptance criterion for the lot.

No submission from this question but it is critical that you understand the issues that such questions raise.

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